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%John Bartlett, jdb382
%Homework Problem #2
%Create a simple animation of interesting motion
%10/9/2017
%%
close all
%Set time for running animation
time = 40;
t = linspace(0,time,1000);

%Set equations of motion
x = @(t) 11*cos(t) - 6*cos(11*t/6);
y = @(t) 11*sin(t) - 6*sin(11*t/6);
theta = @(t) pi/2+atan2(y(t),x(t));
rot = @(t) [cos(theta(t)) sin(theta(t));
            -sin(theta(t)) cos(theta(t))];

%Plot path of object
figure
hold on
axis([-20 20 -20 20]);
plot(x(t), y(t), 'r');

%Create patch object
x_cor = [-.5 -.5;-.5 0;.5 .5;.5 -.5];
y_cor = [-.5 .5;.5 1;.5 .5;-.5 .5];
c = [0; 1];

%%
%Animation Scheme 1: Plot-pause-plot
%patch vertices assigned as object property
axis([-20 20 -20 20]);
obj = patch(x_cor,y_cor,c);
original = obj.Vertices;
for currTime = t
    hold on
    plot(x(t), y(t), 'r'); %Plot path of object
    obj.Vertices = (rot(currTime)*original)';
    trans = [ones(8,1)*x(currTime), ones(8,1)*y(currTime)];
    obj.Vertices = obj.Vertices + trans;
    pause(.01);
    hold off
end

%%
%Animation Scheme 2: Tic-toc & Pause
%patch vertices assigned via setVertices method
figure; hold on; axis([-20 20 -20 20]);
obj = patch(x_cor,y_cor,c);
original = get(obj, 'Vertices');
plot(x(t), y(t), 'r'); %Plot path of object
currTime = 0;
tic
while currTime<time
    set(obj, 'Vertices', (rot(currTime)*original)');
    trans = [ones(8,1)*x(currTime), ones(8,1)*y(currTime)];

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        set(obj,'Vertices', obj.Vertices + trans);
        pause(.01)
        currTime = toc;
end

%%
%Animation Scheme 3: Tic-toc & Drawnow
%patch vertices assigned as object property
figure; hold on; axis([-20 20 -20 20]);
obj = patch(x_cor,y_cor,c);
original = obj.Vertices;
plot(x(t), y(t),'r'); %Plot path of object
currTime = 0;
tic
while currTime<time
    obj.Vertices = (rot(currTime)*original)';
    trans = [ones(8,1)*x(currTime), ones(8,1)*y(currTime)];
    obj.Vertices = obj.Vertices + trans;
    drawnow;
    currTime = toc;
end

%%
%Animation Scheme 4: Animated Line
%not for use in full animation, but can still show motion of an object
figure; hold on; axis([-20 20 -20 20]);
h = animatedline;
currTime = 0;
tic
while currTime<time
    addpoints(h,x(currTime),y(currTime))
    drawnow
    currTime = toc;
end

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